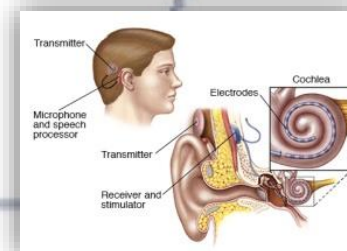
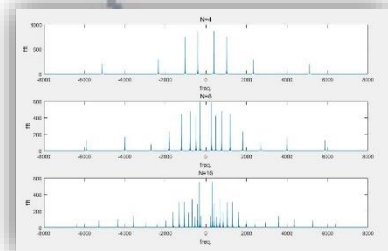
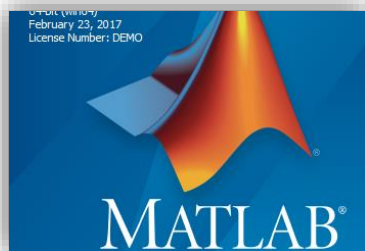


信号与系统实验

主讲人：吴光 博士

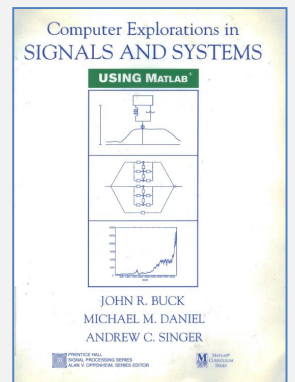
Email: wug@sustech.edu.cn



Labs in this course

5 Lab assignments+2 Projects

- Lab 1: MATLAB Programming (3 Weeks)
- **Lab 2: Linear Time-Invariant Systems (2 Weeks)**
- Lab 3: Fourier Series Representation of Periodic Signals (2 Weeks)
- Lab 4: The Continuous-Time Fourier Transform (2 Weeks)
- Lab 5: Coding Test & Low-Pass Filter (1 Week)
- Project 1: Motion detection via communication signals (3 Weeks)
- Project 2: Modulation Classification with Deep Learning (3 Weeks)



1.1 How to calculate the output of the system ?

- According to the properties of the system, e.g. Linearity, Time-Invariance.
- Step 1: to measure the system function:



- Step 2: to represent the $x(t)$ as a combination of the $\delta(t)/\delta(t - kT_s)$:



Requirement: H is a LTI system

2.2 Impulse Response

- Particularly, $y[n]$ will be impulse response as the input is impulse function
 - $y[n] = 0.5x[n] + x[n - 1] + 2x[n - 2]$
 $h[n] = ?$ Finite Impulse Response (FIR)
 - $y[n] - 0.8y[n - 1] = 2x[n]$
 $h[n] = ?$ Infinite Impulse Response (IIR)
- **Causal** DT LTI system is uniquely specified by two vectors: **$A = [a_0 \ a_1 \ a_2 \ \dots \ a_K]$** and **$B = [b_0 \ b_1 \ \dots \ b_M]$**
 - $A = [1], \ B = [0.5 \ 1 \ 2]$
 - $A = [1 \ -0.8], \ B = [2]$

$$\sum_{k=0}^K a_k y[n - k] = \sum_{m=0}^M b_m x[n - m]$$

2.4 conv and filter

	conv	filter
Scenario	• Finite signal • Finite impulse response	• Finite signal • Infinite/finite impulse response
Remarks	• $nw = nu(1) + nv(1) : nu(end) + nv(end)$	• x and y share the same range of time indices • Truncated output

Signals and Systems (Lab)

Lab 2: Linear Time-Invariant Systems

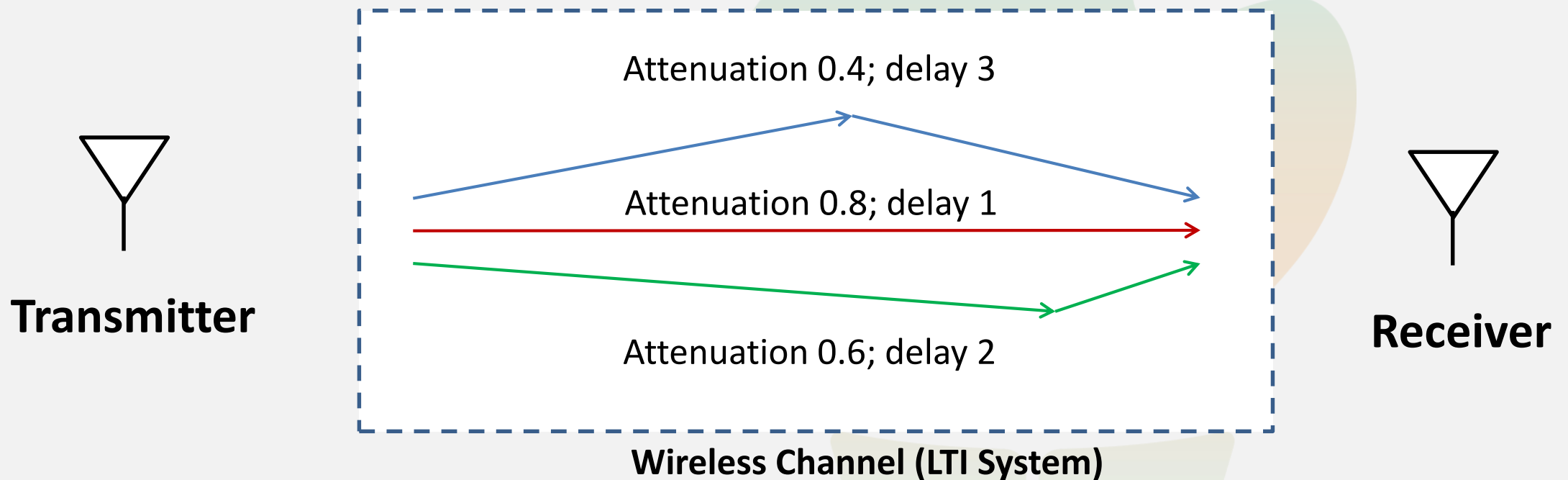
Part 2

Dr. **Wu Guang**

wug@sustech.edu.cn

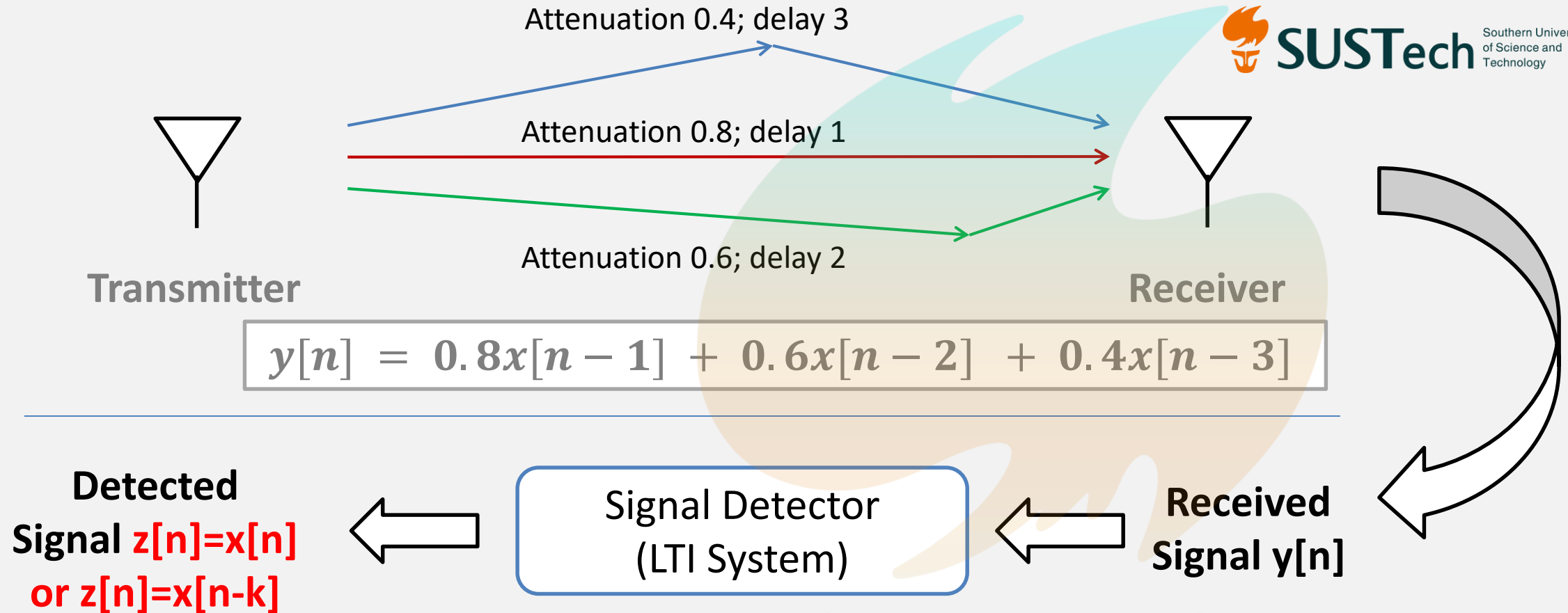
Electrical & Electronic Engineering
Southern University of Science and Technology

3.1 Application: Simplified Wireless Channel



Impulse response: $h_1[n] = 0.8\delta[n-1] + 0.6\delta[n-2] + 0.4\delta[n-3]$

Difference equation: $y[n] = 0.8x[n-1] + 0.6x[n-2] + 0.4x[n-3]$



Impulse response: $h2[n] * h1[n] = \delta[n] \text{ or } \delta[n-k]$

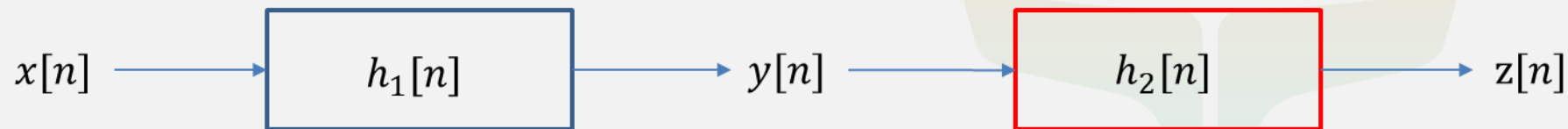
Difference Equation: $0.8z[n-1] + 0.6z[n-2] + 0.4z[n-3] = y[n]$

$$z'[n] = z[n-1] \Rightarrow 0.8z'[n] + 0.6z'[n-1] + 0.4z'[n-2] = y[n]$$

3.2 Properties of Convolution

$$y[n] = x[n] * h_1[n]$$

$$z[n] = y[n] * h_2[n] = x[n] * \underbrace{h_1[n] * h_2[n]}_{\delta[n]} \rightarrow x[n]$$



3.2 Properties of Convolution

$$h_1[n] = 0.8\delta[n - 1] + 0.6\delta[n - 2] + 0.4\delta[n - 3]$$

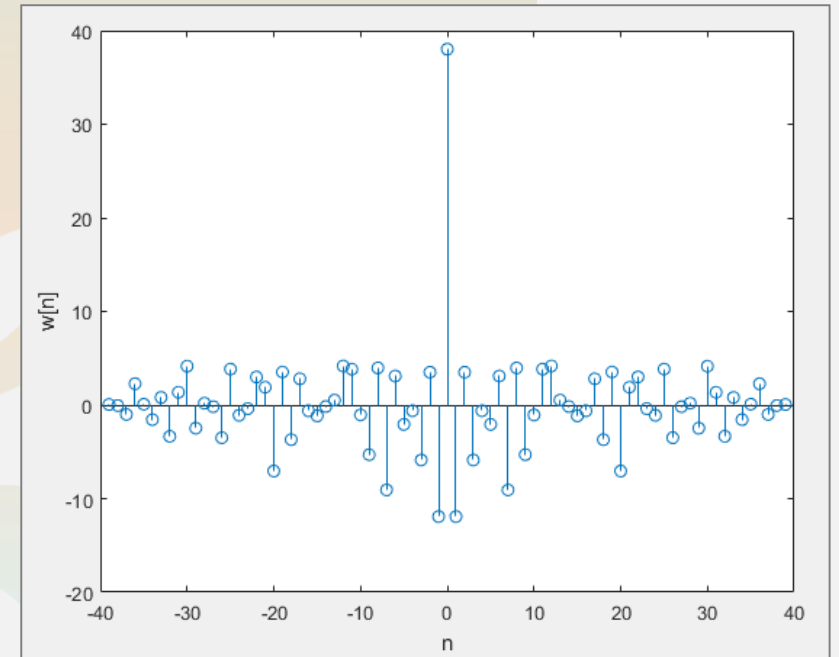
$$h_2[n] * h_1[n] = h_2[n] * (0.8\delta[n - 1] + 0.6\delta[n - 2] + 0.4\delta[n - 3])$$

$$= 0.8h_2[n - 1] + 0.6h_2[n - 2] + 0.4h_2[n - 3] = \delta[n]$$

4.1 Signal Auto-Correlation

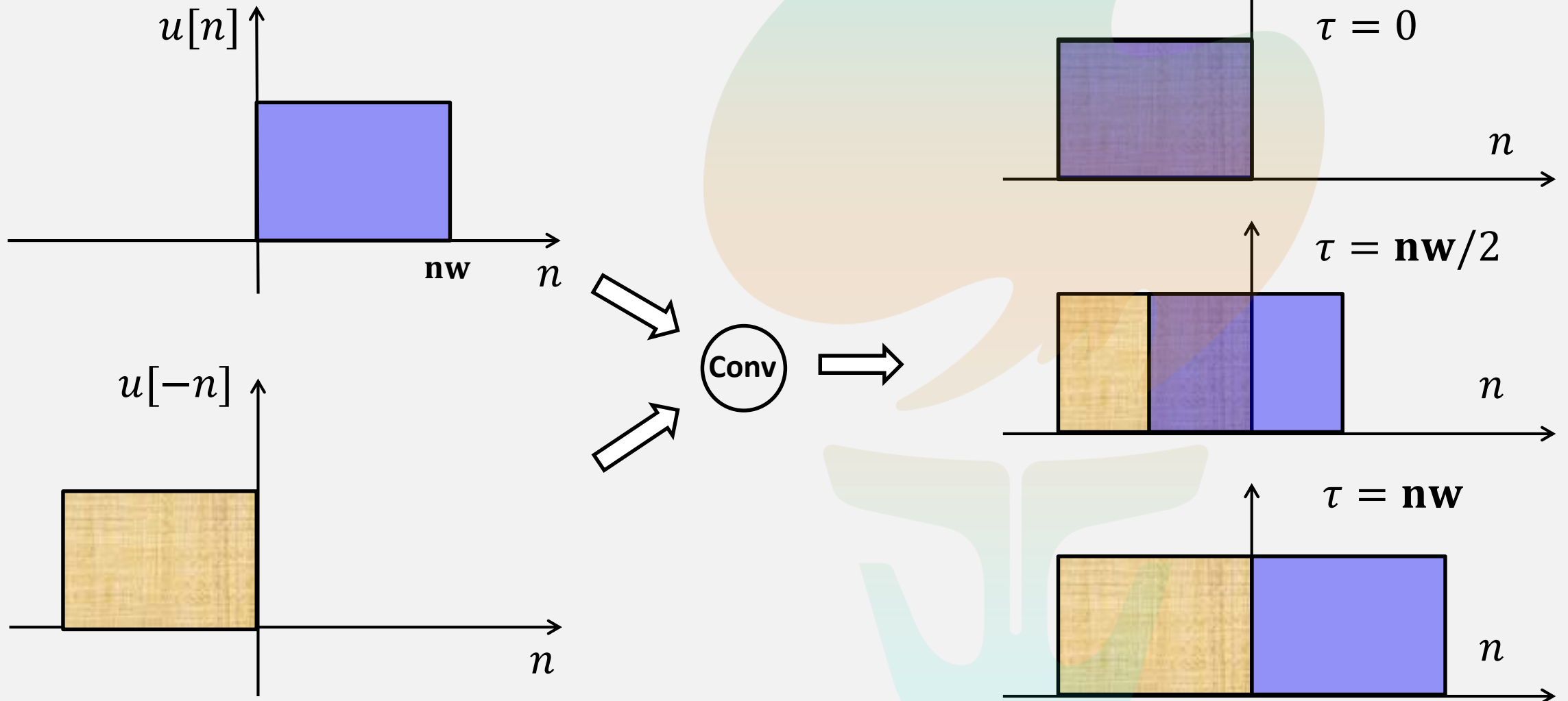
- Auto-correlation of $u[n]$: $w[n] = u[n] * u[-n]$

```
u=randn(1,40);  
nu = 1:40;  
v=u(end:-1:1);  
nv=-40:-1;  
w=conv(u,v);  
nw=nu(1)+nv(1):nu(end)+nv(end);  
stem(nw,w)
```

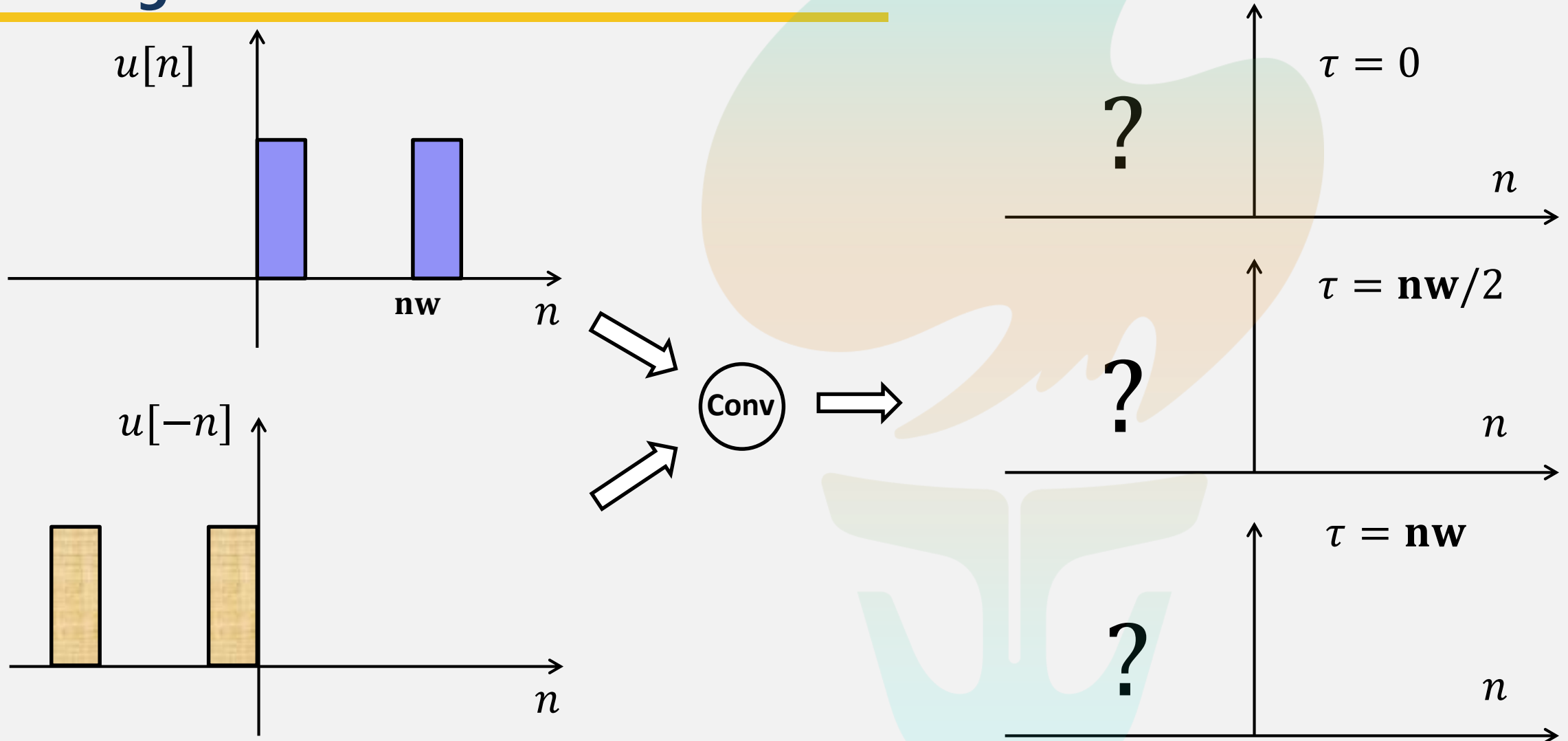


Auto-correlation of a random signal has a high peak at the origin

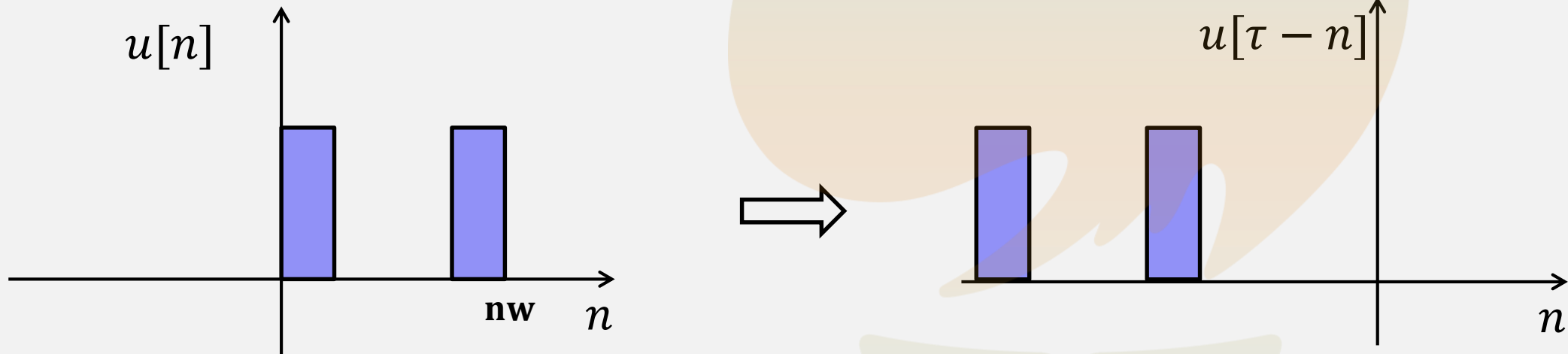
4.2 Signal Auto-Correlation

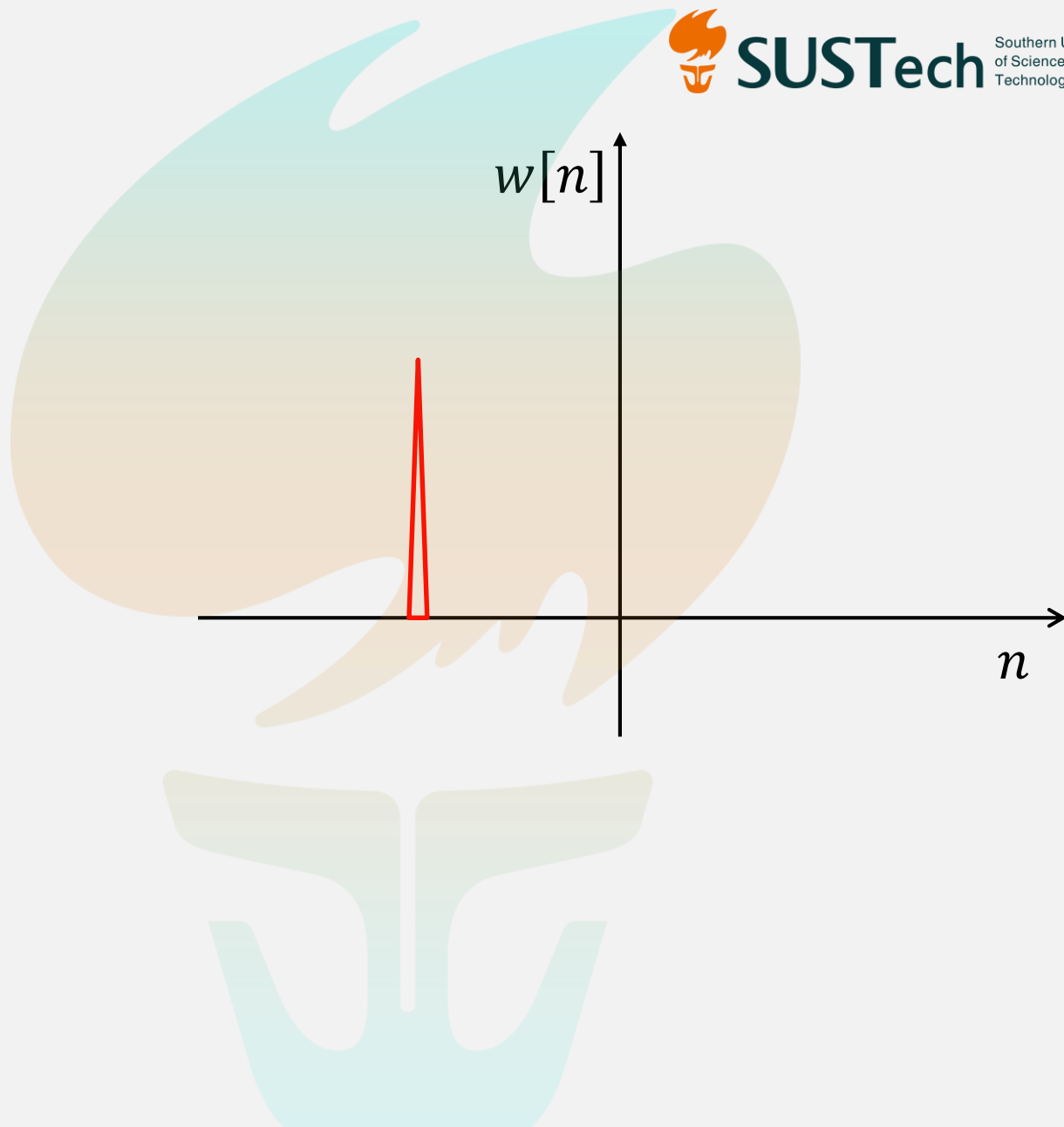
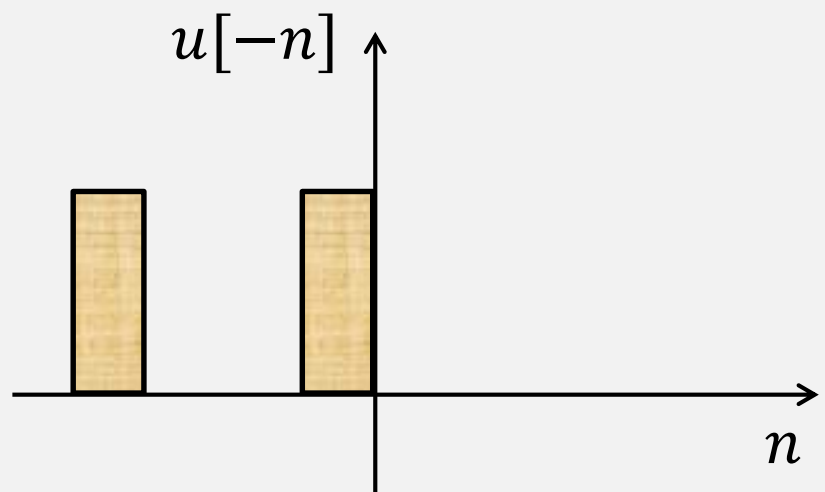
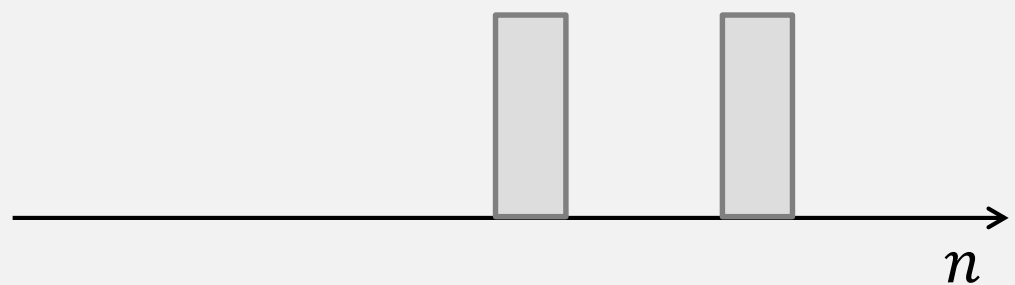
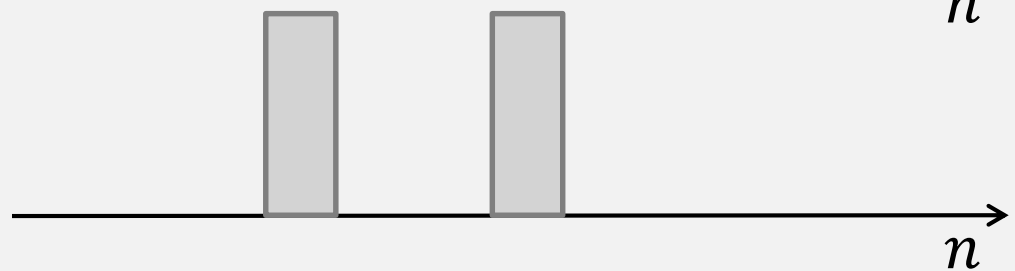
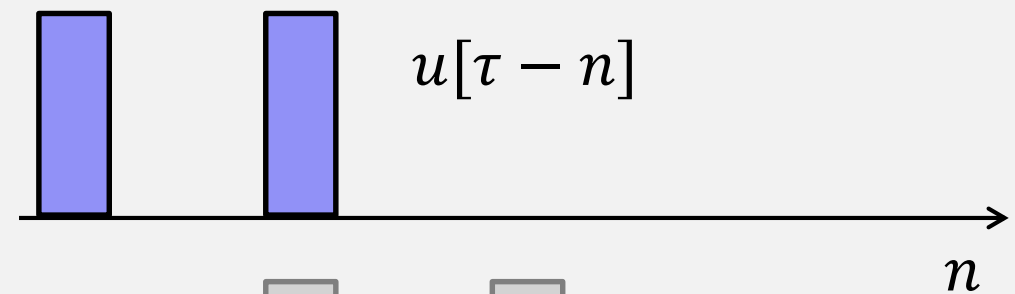


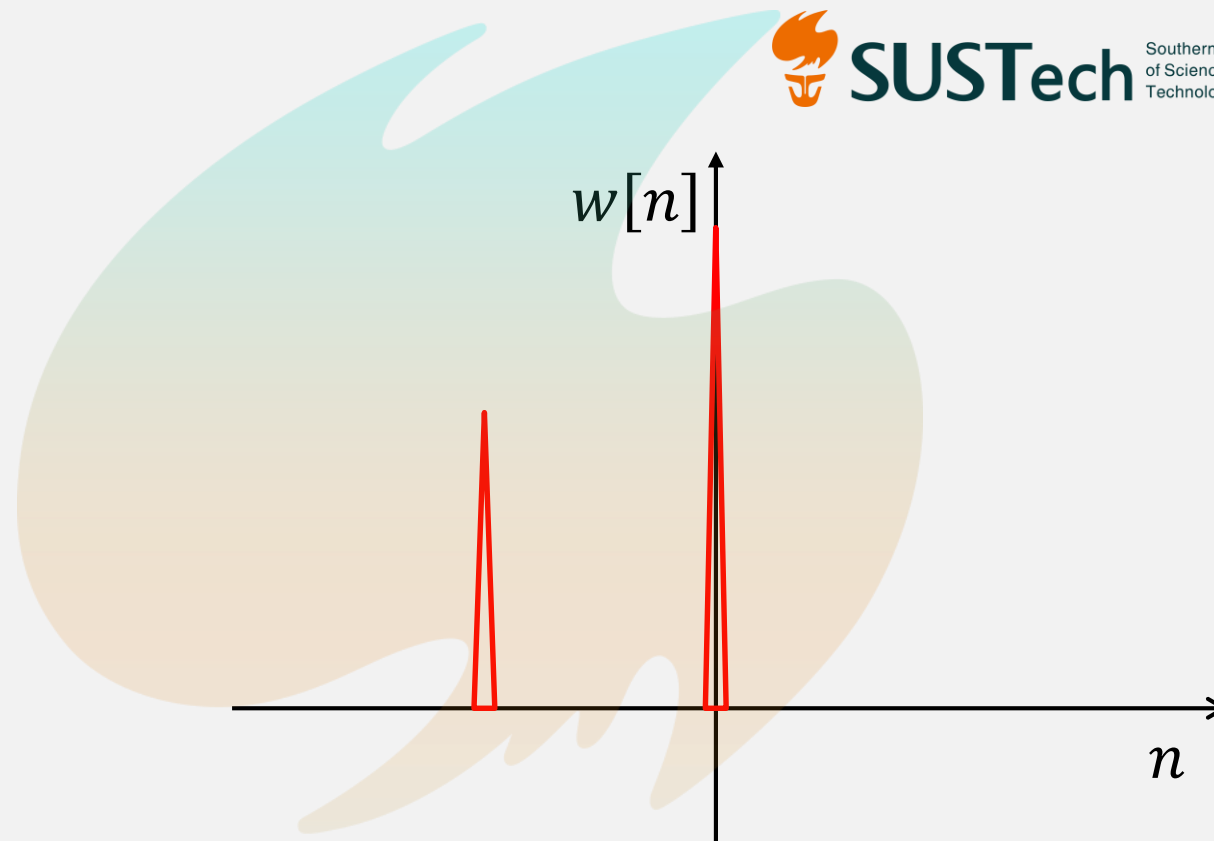
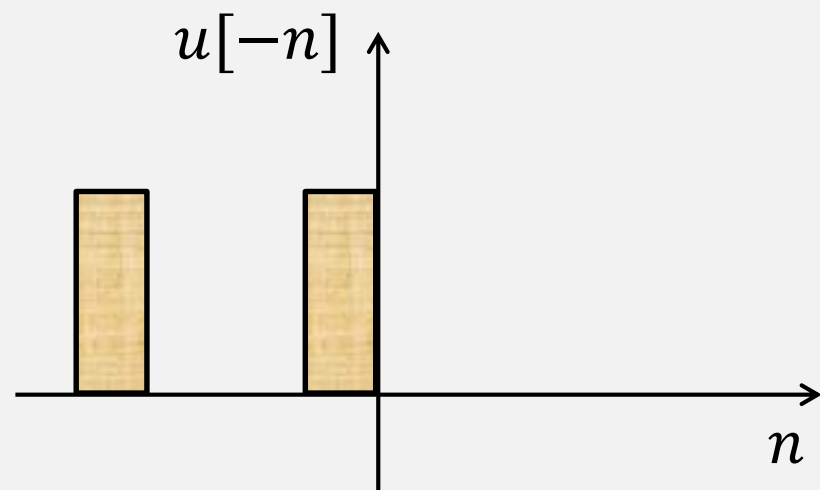
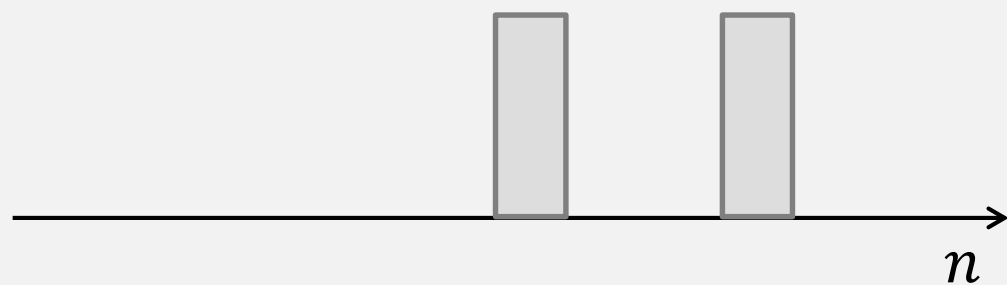
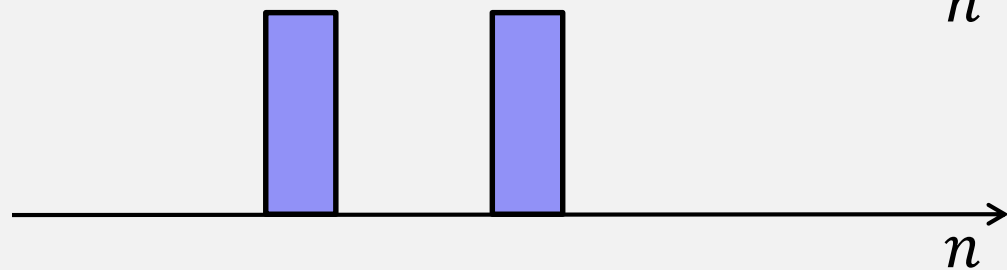
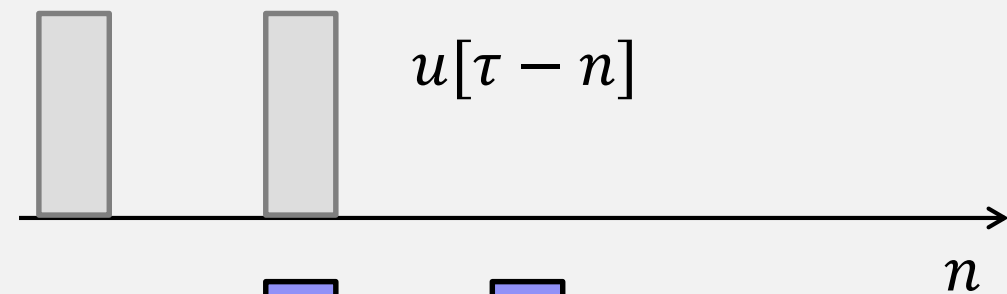
4.3 Signal Auto-Correlation

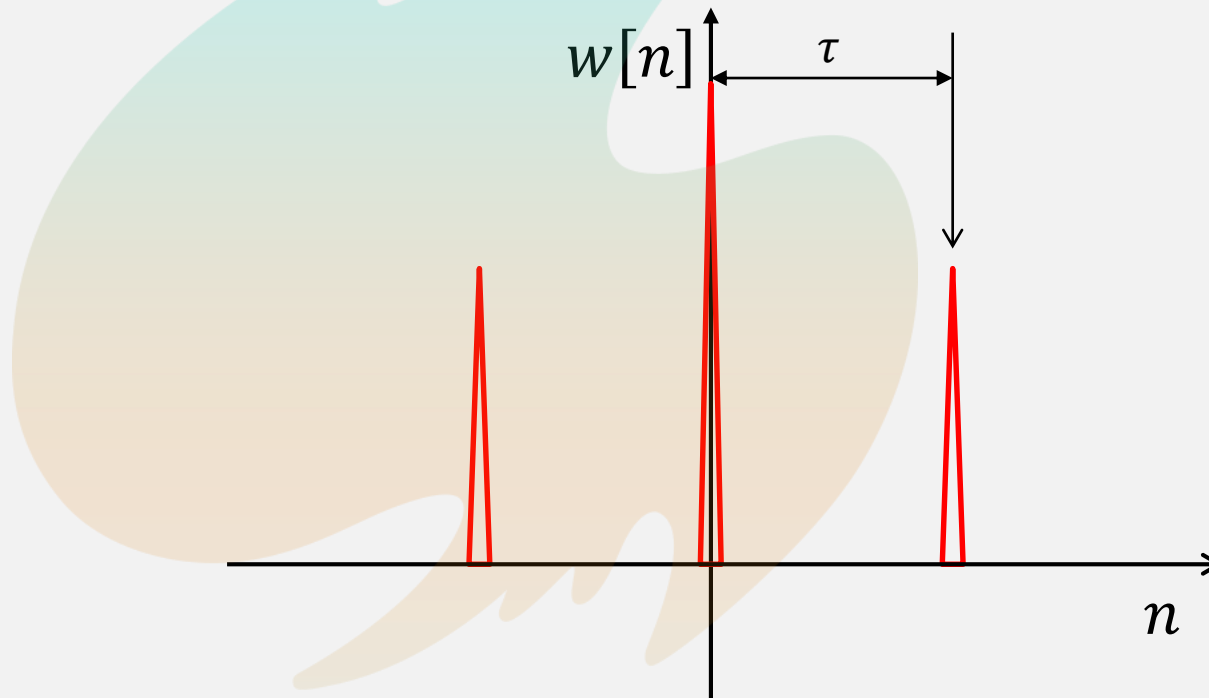
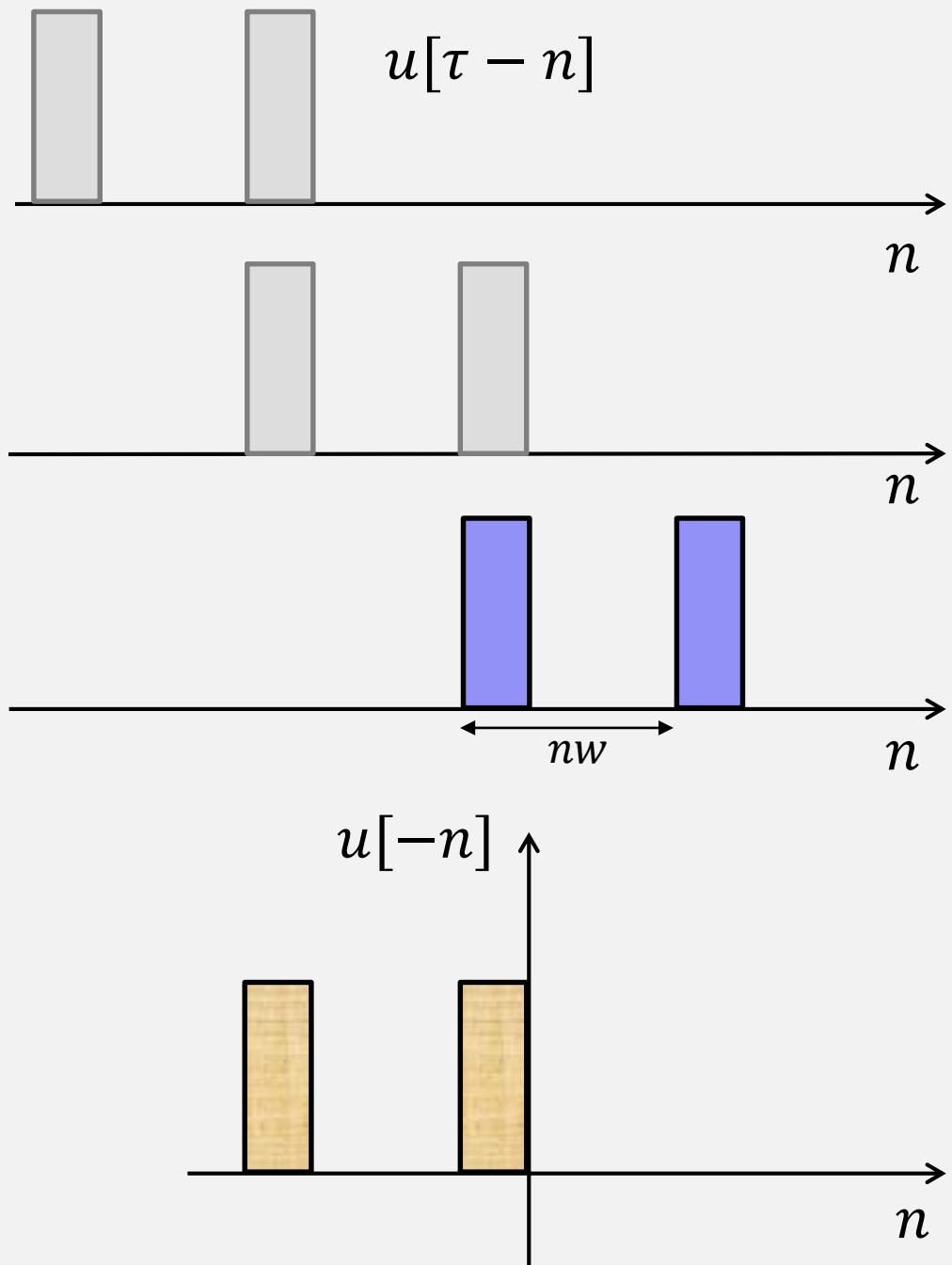


4.3 Signal Auto-Correlation



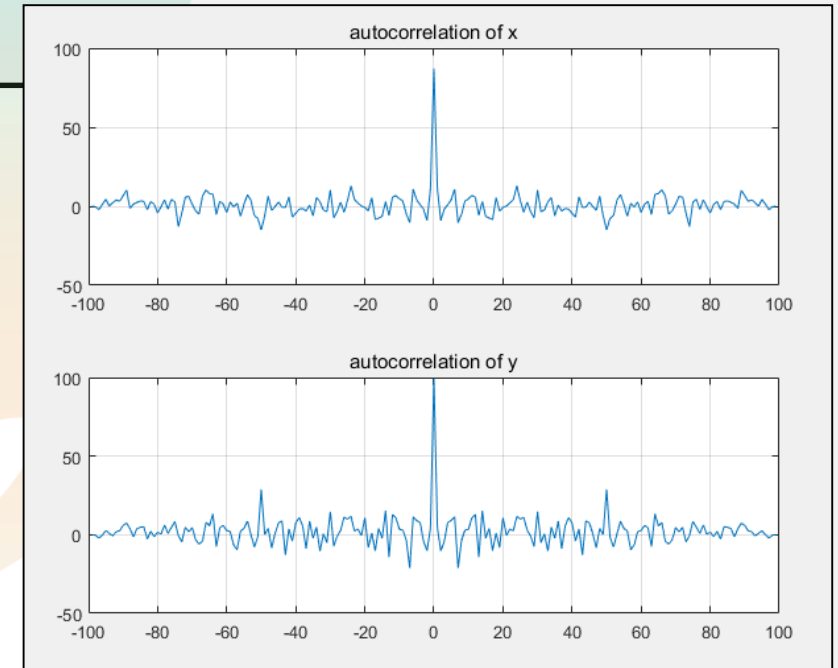






4.4 Exercise 4

```
NX = 100;  
x = randn(1,NX);  
N = 50;  
alpha = 0.9;  
yex = filter([1,zeros(1,N-1),alpha],1,x);  
Rxx = conv(x,flipr(x));  
Ryy = conv(yex,flipr(yex));  
figure;subplot(212);  
plot([-NX+1:NX-1],Ryy); grid on; title('autocorrelation of y');  
subplot(211);  
plot([-NX+1:NX-1],Rxx); grid on; title('autocorrelation of x');
```



3.2 Application: Echo-Time Estimation

Suppose that you were given $y[n]$ but did not know the value of the echo time, N , or the amplitude of the echo, α . Based on Eq. (2.21), can you determine a method of estimating these values? Hint: Consider the output y of the echo system to be of the form:

$$y[n] = x[n] * (\delta[n] + \alpha\delta[n - N])$$

and consider the signal,

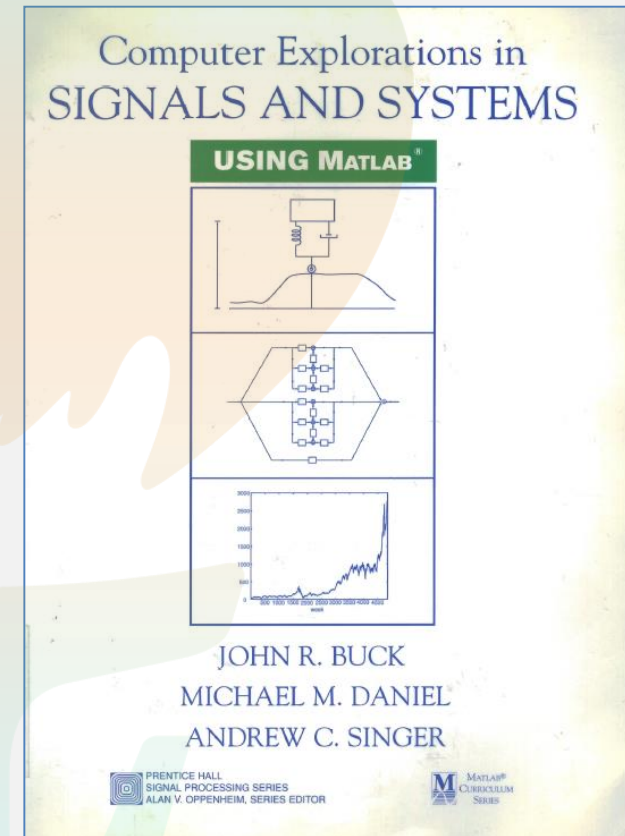
$$R_{yy}[n] = y[n] * y[-n].$$

This is called the autocorrelation of the signal $y[n]$ and is often used in applications of echo-time estimation. Write $R_{yy}[n]$ in terms of $R_{xx}[n]$ and also plot $R_{yy}[n]$. You will

$$\begin{aligned} R_{yy}[n] &= x[n] * (\delta[n] + \alpha\delta[n - N]) * x[-n] * (\delta[n] + \alpha\delta[n + N]) \\ &= R_{xx}[n] * ((1 + \alpha^2)\delta[n] + \alpha\delta[n - N] + \alpha\delta[n + N]) \end{aligned}$$

Lab Assignment

- Read tutorial 2.1 & 2.2 & 2.3 & 2.4 by yourself
- 2.10
 - The sound file for 2.10: **lineup.mat**
- Submit your report.

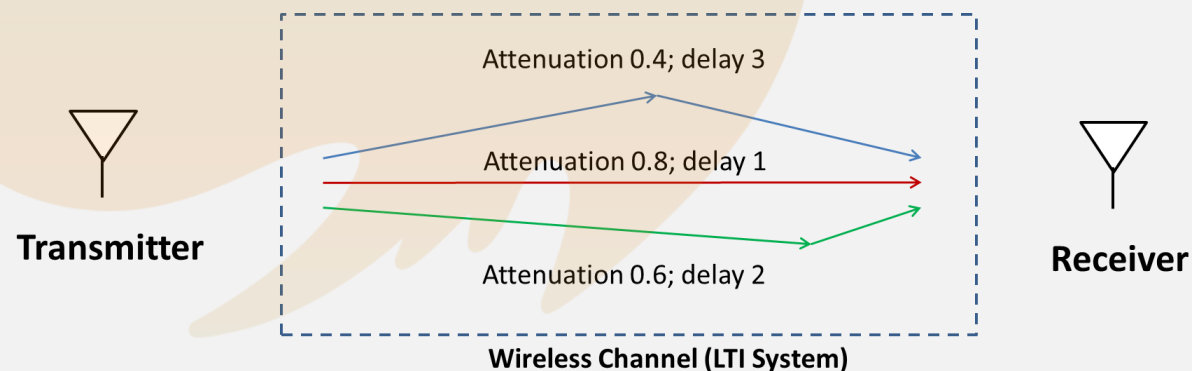


Coding Test – (30 mins)

巴克码序列因其具有强自相关性，被应用于当前主流无线通信系统中，例如WiFi系统，如图(a)所示。设无线信道由图(b)所示的LTI系统描述，

Code Length	Barker Sequence
2	[-+, --]
3	[- - +]
4	[- + --, - + ++]
5	[- - - + -]
7	[- - - + + - +]
11	[- - - + + + - + + - +]
13	[- - - - - + + - - + - + -]

(a)



(b)

- 1、设发射机发射的信号为长度11的巴克码序列，计算接收机接收的信号，画出结果图。
- 2、设计逆系统，从接收机接收的信号中恢复巴克码序列，画出结果图。
- 3、计算长度为7/11/13的巴克码序列与自身镜像的卷积，画出结果图。

Question ?

